

NIKOLAI, AK, USA

WMO: 702625

Lat: 63.017N

Lon: 154.367W

Elev: 126

StdP: 99.82

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00104

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.6	-38.8	-37.2	0.1	-33.2	-35.9	0.1	-32.7	10.2	-13.0	9.1	-12.1	0.1	270	0.700

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	26.4	15.8	23.8	14.5	22.0	13.6	16.7	23.8	15.6	21.6	14.7	20.0	2.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	10.2	17.7	13.0	9.5	16.9	12.3	9.1	16.5	47.0	23.5	44.0	21.4	41.3	20.1	20.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.3	6.2	DB	-43.2	29.6	4.0	2.3	-46.1	31.3	-48.5	32.6	-50.7	33.9	-53.6	35.6	(4)
(5)			WB	-34.9	17.7	3.9	1.7	-37.7	18.9	-40.0	19.9	-42.2	20.9	-45.1	22.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.5	-19.1	-13.8	-9.7	0.4	8.7	13.7	15.4	12.4	7.2	-1.7	-14.6	-17.6	(6)	
	DBStd	14.22	11.44	8.37	7.66	5.80	4.22	2.97	3.07	3.06	3.89	6.19	9.01	9.91	(7)	
	HDD10.0	4607	902	666	611	289	73	3	1	10	96	363	737	856	(8)	
	HDD18.3	7253	1161	899	869	539	300	143	102	185	333	621	987	1114	(9)	
	CDD10.0	409	0	0	0	0	32	113	167	84	13	1	0	0	(10)	
	CDD18.3	15	0	0	0	0	0	4	9	1	0	0	0	0	(11)	
	CDH23.3	242	0	0	0	0	8	83	133	18	0	0	0	0	(12)	
	CDH26.7	51	0	0	0	0	1	23	24	2	0	0	0	0	(13)	
(14) Wind	WSAvg	2.1	2.2	2.4	2.5	2.7	2.3	2.0	1.9	1.8	1.9	2.1	1.9	1.9	(14)	
(15) Precipitation	PrecAvg	591	40	36	29	23	30	50	72	84	74	57	49	48	(15)	
	PrecMax	743	77	98	69	46	57	95	143	133	144	164	132	100	(16)	
	PrecMin	444	6	3	2	6	10	20	38	27	27	12	4	10	(17)	
	PrecStd	80	20	19	16	10	12	17	23	23	35	30	32	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	4.0	8.7	16.3	25.2	29.9	29.4	25.8	19.9	13.8	6.1	5.2	(19)	
		MCWB	1.2	0.3	3.3	7.2	13.4	15.5	17.7	16.3	12.2	7.6	4.6	0.8	(20)	
	2%	DB	1.8	1.6	5.1	13.4	22.3	25.8	27.0	22.6	17.5	10.1	2.2	0.5	(21)	
		MCWB	-0.2	-0.4	1.4	5.6	11.9	15.0	16.2	14.8	10.9	6.1	1.0	-0.8	(22)	
	5%	DB	-1.3	-0.1	2.8	11.7	19.5	22.9	24.5	20.2	15.7	7.7	-0.1	-2.5	(23)	
		MCWB	-2.5	-1.5	-0.3	5.0	10.7	13.2	15.8	13.8	10.3	5.0	-1.1	-3.6	(24)	
	10%	DB	-5.0	-2.6	0.9	9.0	17.2	21.1	22.4	18.0	13.3	5.9	-2.8	-5.0	(25)	
		MCWB	-6.1	-4.0	-1.1	3.6	9.2	12.6	14.7	12.7	9.4	4.1	-3.7	-5.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	1.5	3.5	7.7	13.8	17.4	18.9	17.0	12.8	8.2	4.1	1.8	(27)	
		MCDB	4.7	3.5	7.5	15.6	23.6	25.9	26.6	23.6	17.5	12.2	6.9	4.7	(28)	
	2%	WB	0.1	-0.1	1.7	6.1	12.3	15.6	17.3	15.7	11.8	6.6	1.0	-0.6	(29)	
		MCDB	2.1	1.4	4.6	12.4	21.2	23.1	25.5	20.3	15.9	9.4	2.0	1.0	(30)	
	5%	WB	-2.1	-1.7	0.2	5.1	10.9	14.3	16.3	14.6	10.8	5.3	-1.2	-3.1	(31)	
		MCDB	-0.8	0.0	2.5	11.0	18.3	21.0	22.8	18.5	14.1	7.5	0.0	-2.3	(32)	
	10%	WB	-5.1	-3.9	-1.4	3.9	9.9	13.2	15.3	13.7	10.0	4.1	-3.7	-5.2	(33)	
		MCDB	-3.9	-2.5	1.0	9.0	16.1	19.3	21.0	17.2	13.3	6.0	-2.7	-4.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	9.4	11.6	11.1	11.8	10.8	10.0	9.0	8.6	6.5	7.6	7.6	(35)	
		MCDBR	8.2	9.3	11.8	13.9	16.1	14.8	14.5	12.6	12.7	8.5	7.2	9.4	(36)	
	5% WB	MCWBR	6.7	7.9	8.6	7.9	7.7	6.1	6.2	5.8	7.3	5.6	6.0	8.2	(37)	
		MCWBR	8.4	8.8	9.6	13.1	14.2	12.8	13.2	9.8	10.3	7.5	7.2	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.199	0.256	0.272	0.323	0.362	0.377	0.377	0.382	0.310	0.282	0.213	0.191	(40)	
		taud	2.024	2.231	2.272	2.245	2.365	2.341	2.385	2.377	2.558	2.454	2.022	1.911	(41)	
	Ebn at Noon		457	702	840	857	853	843	833	791	804	680	409	223	(42)	
		Edh at Noon	31	59	83	108	105	110	103	96	65	48	30	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.25	0.97	2.77	4.60	5.16	5.31	4.55	3.48	2.29	0.98	0.33	0.10	(44)	
	RadStd		0.03	0.09	0.25	0.34	0.49	0.36	0.40	0.35	0.32	0.12	0.05	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+1.00	N/A	+0.85	N/A	N/A	N/A	-299	-358	N/A	N/A

Nomenclature: See separate page